Failure Analysis Report

FAIL-0349 — Synchronization Failure

1. Failure Summary

Failure ID	FAIL-0349
Failure Type	Synchronization Failure
Severity	Minor
Component	Payment Gateway
Application	APP-008 — VoiceAssistant
Build Version	5.1.5
Timestamp	2024-11-05 08:49:48
Environment	ENV-010 — High Load Env
OS Version	Ubuntu 22.04
Network	Ethernet
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-0926
Title	Verify that push notification delivery fails gracefully with a user-friendly err...
Execution ID	EXEC-01713
Test Run	RUN-009 — Regression Cycle 9 — Build 6.1.8
Requirement	REQ-0131 — Data Synchronization — Search Query Execution

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete Bluetooth reconnection within the defined SLA.
Observed Behaviour	The system exhibited synchronization failure when the device has been offline for more than 24 hours...

4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-8589'; last_modified timestamps diverge by 94s.
```

Stack Trace / Log Excerpt:

```
com.samsung.smarthome.PaymentGateway.execute(line 554)
```

Log Reference	LOG-68069

5. Executed Steps at Time of Failure

1	Observe the status indicator for up to 10 seconds.	A success toast message is displayed.

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2	Navigate to Settings > Account and tap 'Pair Device'.	The fallback UI is displayed with a human-readable error message.
3	Navigate to Settings > Bluetooth and tap 'Pair Device'.	The audit log records the event with correct metadata.

6. Root Cause Analysis

Race condition in the connection manager when two threads attempt simultaneous reconnection.

7. Linked Defect

Defect ID	BUG-0047
Title	Synchronization Failure in Payment Gateway: SyncError: conflict detected on enti...
Severity	Minor
Priority	Critical
Status	REOPENED
Root Cause	Race condition in the connection manager when two threads attempt simultaneous r...
Resolution	—
Assigned To	ENG-072

8. Historical Similar Failures

HIST-0348	FAIL-0349	Synchronization Failure	6.0.1	2023-05-18	7	Resolved
HIST-0524	FAIL-0525	Synchronization Failure	3.4.0	2023-12-20	1	Wont Fix
HIST-0593	FAIL-0594	Synchronization Failure	5.0.6	2024-01-04	10	Wont Fix

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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